2.5 Constructing Parallel & Perpendicular

Compass-and-straightedge constructions produce results that are exactly correct, provable through triangle congruence, rather than approximately correct from a measured reading. This section pushes past memorizing steps by asking *why* each construction works — usually because two equal-radius arcs force an SSS congruence, which in turn forces a right angle or equal segment. It also chains the corresponding-angle idea from Section 2.2 into an actual construction for a parallel line, and applies the perpendicular bisector to a genuine problem: locating the center of a circle from three points on it.

Examples

Example 1: Tools of a construction

What two tools are allowed in a classical geometric construction, and why are a ruler's numbered markings and a protractor's degree markings both excluded?

Solution

The two allowed tools are a compass (draws circles and arcs of a fixed radius) and an unmarked straightedge (draws straight lines, but does not measure).

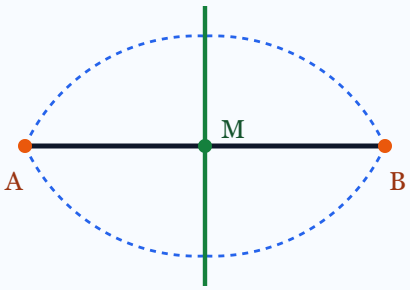
Numbered rulers and protractors are excluded because a construction must be logically exact — reading "5.0 cm" or "37^\circ" off a scale is only as good as the tool's smallest marking, an approximation.

Answer: Compass and straightedge only, because they reproduce exact equal lengths and equal angles without relying on any measurement.

Example 2: Perpendicular at a point on a line

Describe the steps to construct a perpendicular to line ℓ at a point P on ℓ, and explain why the result is guaranteed to be exactly $90°$.

Solution



Equal arcs from two equidistant points meet above and below, pinning down a perpendicular.

Steps: (1) center the compass at P and draw arcs of equal radius crossing ℓ on both sides, at A and B; (2) from A and from B, draw two more equal-radius arcs (wider than half AB) that intersect above P, at Q; (3) draw $\overrightarrow{PQ}$.

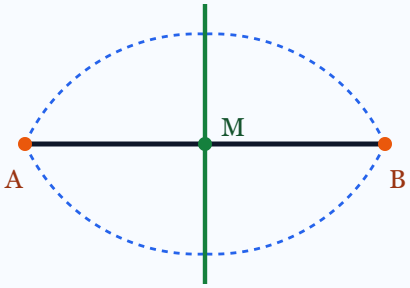
Reasoning: $PA = PB$ (first radius) and $QA = QB$ (second radius), with $PQ = PQ$ shared, so $\triangle QPA \cong \triangle QPB$ by SSS — their angles at P are congruent and form a linear pair, forcing each to be $90°$.

Answer: $\overrightarrow{PQ} \perp \ell$, guaranteed exact by SSS congruence, not by measuring.

Example 3: Perpendicular bisector locates the midpoint too

Construct the perpendicular bisector of $\overline{AB}$, and explain why the point where it crosses $\overline{AB}$ is exactly the midpoint.

Solution



The same construction proves perpendicularity and the midpoint at once.

Steps: from A and separately from B, draw arcs of equal radius (more than half AB) intersecting above and below the segment at P and Q; draw $\overleftrightarrow{PQ}$.

Since $AP = BP$ and $AQ = BQ$ (equal compass radii), with $PQ = PQ$ shared, $\triangle APQ \cong \triangle BPQ$ by SSS, forcing $\angle APQ \cong \angle BPQ$ — a congruent linear pair, so each is $90°$; CPCTC then gives $AM = BM$ at the crossing point M.

Answer: $\overleftrightarrow{PQ}$ is perpendicular to $\overline{AB}$ at its midpoint M — one construction proves both facts.

Example 4: Perpendicular from a point not on the line

Describe the steps to drop a perpendicular from a point P (not on line ℓ) to ℓ.

Solution

Steps: (1) from P, draw an arc that crosses ℓ at two points, A and B; (2) from A and from B, draw two more equal-radius arcs on the opposite side of ℓ from P that intersect at Q; (3) draw $\overleftrightarrow{PQ}$.

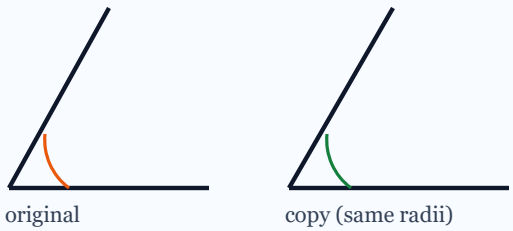
Reasoning: $PA = PB$ from step 1 and $QA = QB$ from step 2, so both P and Q are equidistant from A and B — meaning $\overleftrightarrow{PQ}$ is the perpendicular bisector of $\overline{AB}$, which is therefore perpendicular to ℓ (since $A, B \in \ell$).

Answer: $\overleftrightarrow{PQ}$ meets ℓ at a right angle — the foot of the perpendicular from P.

Example 5: Copying an angle

Describe how to copy $\angle A$ onto a new ray using only compass and straightedge, and explain why the copy is exactly congruent.

Solution



Matching all three sides of the small triangle forces matching angles.

Steps: (1) draw an arc centered at the vertex of $\angle A$, crossing both rays at B and C; (2) draw a new ray, and with the *same* compass radius, draw an arc from the new vertex crossing the new ray at B'; (3) open the compass to width BC, and from B' draw an arc crossing the first new arc at C'; (4) draw the ray through C'.

Reasoning: the new triangle (vertex, B', C') has all three sides equal to the original (vertex, B, C) triangle by construction, so they are congruent by SSS, forcing the angles at the vertices to match.

Answer: The new angle is exactly congruent to $\angle A$, guaranteed by SSS — not by eyeballing it.

Example 6: Constructing a parallel line

Explain how copying a corresponding angle at a new point P lets you construct a line through P parallel to a given line ℓ.

Solution

Draw a transversal through P that also crosses ℓ, creating an angle at ℓ. Using the Example 5 method, copy that angle at P, in the matching corresponding position along the same transversal.

The new angle is congruent to the original by construction, so the two lines now have congruent corresponding angles with the same transversal.

By the Converse of the Corresponding Angles Postulate (Section 2.2), congruent corresponding angles force the lines to be parallel.

Answer: The newly drawn line through P is parallel to ℓ.

Example 7: Why constructions outrank measuring

A ruler reads to the nearest millimeter and a protractor to the nearest degree. Explain why a compass-and-straightedge construction counts as more 'exact,' even though a compass is a physical tool too.

Solution

A ruler or protractor reading is inherently an approximation — the true value could fall anywhere within the tool's smallest marked interval.

A compass construction instead relies on logical guarantees, like SSS congruence, that hold exactly regardless of how carefully the compass was drawn — the *proof* guarantees the result, not the drawing's precision.

Answer: Constructions are exact because their correctness is proved through congruence, not read off a scale.

Example 8: Uniqueness of the perpendicular

Through a given point, how many lines can be drawn perpendicular to a given line? Justify your answer using the constructions above.

Solution

Both perpendicular constructions above (Examples 2 and 4) always produce one specific line — nothing in the steps allows a second, different perpendicular through the same point.

This matches the Uniqueness of Perpendiculars Theorem.

Answer: Exactly one perpendicular line through a given point.

Example 9: Uniqueness of the parallel

Through a point P not on line ℓ, how many lines can be drawn parallel to ℓ in Euclidean geometry?

Solution

The construction in Example 6 produces exactly one specific parallel line through P.

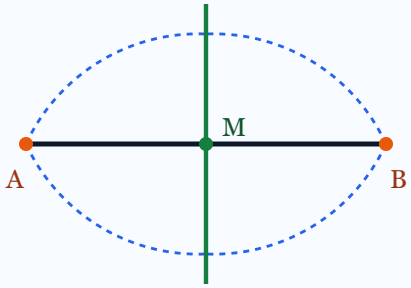
Euclidean geometry's Parallel Postulate (Playfair's form) guarantees that no second, different parallel line through P exists.

Answer: Exactly one — this uniqueness is special to Euclidean geometry.

Example 10: Application — finding a circle's center

Three points A, B, C lie on a circle whose center is unmarked. Explain how two perpendicular-bisector constructions can locate the exact center.

Solution



The center is equidistant from A and B — so it lies on this exact line.

The center of a circle is equidistant from every point on it, so it must be equidistant from A and B — meaning it lies on the perpendicular bisector of $\overline{AB}$.

By the same reasoning, it must also lie on the perpendicular bisector of $\overline{BC}$.

Construct both bisectors (Example 3's method, applied twice); their single intersection point is equidistant from all three points, so it must be the center.

Answer: The center is the intersection of the perpendicular bisectors of $\overline{AB}$ and $\overline{BC}$.

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Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Tools

Two tools for constructions?

Answer: **Verification:**

Problem 2: Protractor?

Is a protractor allowed in constructions?

Answer: **Verification:**

Problem 3: Perp at on-line point

Construction of ⊥ at a point on a line uses equal what?

Answer: **Verification:**

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Problem 4: Perp from off-line point

This construction creates a line from the point that is what to the given line?

Answer: **Verification:**

Problem 5: Parallel uniqueness

Parallels through an external point: how many?

Answer: **Verification:**

Problem 6: Perp uniqueness

Perpendiculars through a point to a line: how many?

Answer: **Verification:**

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Problem 7: Perp bisector

Perpendicular bisector intersects the segment at its what?

Answer: **Verification:**

Problem 8: Justify parallel

Congruent corresponding angles imply lines are what?

Answer: **Verification:**

Problem 9: Justify perp

If an angle formed by two equal-radius arc constructions is $90°$, the lines are what?

Answer: **Verification:**

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Problem 10: Ruler measure?

May you transfer a length by reading a ruler in a construction?

Answer: **Verification:**

Problem 11: Copy angle

Copying an angle relies on which triangle congruence shortcut?

Answer: **Verification:**

Problem 12: Result

Construction through $P \notin \ell$ parallel to ℓ yields a line that never does what with ℓ?

Answer: **Verification:**

Problem 13: Result

Construction of ⊥ yields what angle measure?

Answer: **Verification:**

Problem 14: Challenge — midpoint without a ruler

A carpenter needs the exact midpoint of a board with no ruler. Which construction should be used, and what two facts does it guarantee simultaneously?

Answer: **Verification:**

Problem 15: Challenge — construct 60°

Explain how to construct a $60°$ angle with no protractor, using only an equilateral triangle built from segment $\overline{AB}$.

Answer: **Verification:**

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✓ Answer Key

1. Compass and straightedge
2. No
3. Equal arcs / equal radii
4. Perpendicular
5. One
6. One
7. Midpoint
8. Parallel
9. Perpendicular
10. No
11. SSS
12. Intersects
13. $90°$
14. The perpendicular bisector construction; it guarantees the point is the exact midpoint AND that the bisector line is perpendicular to the board
15. Draw $\overline{AB}$; with radius AB, draw arcs from A and B meeting at C; since $AB = AC = BC$, $\triangle ABC$ is equilateral, so each angle equals $60°$

Great work — keep practising!